# Practice workbook · Write an algorithm on paper: flowcharts and pseudocode

Producing an algorithm on paper that meets a written requirement list using sequence, selection and iteration: drawing a flowchart, arranging supplied flowchart boxes, or filling in the missing statements of a pseudocode skeleton.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-06

# 1. Question 6 · part a · 6 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part a only.

**6** Algorithms can be represented using flowcharts or pseudocode.

(a) An algorithm has been designed to perform these steps:

- A user is asked to input a number **n**
- If the number **n** is 0 the program stops
- If the number **n** is positive a message to display 'Positive' is output
- If the number **n** is negative a message to display 'Negative' is output
- The program repeats until a value of 0 has been input.

Draw a flowchart to represent this algorithm.

**(6)**

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# 1. Question 6 · part a · 6 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part a only.

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(b) **Figure 3** shows an algorithm written in pseudocode that performs a mathematical process.

```
01
02  RECEIVE number FROM (INTEGER) KEYBOARD
03
04  SET flag TO FALSE
05
06  IF number = 1 THEN
07     SET flag TO TRUE
08  ELSE
09     SET testNum TO 1
10     REPEAT
11        SET testNum TO testNum + 1
12        IF (number MOD testNum) = 0 THEN
13           SET flag TO TRUE
14        END IF
15     UNTIL (testNum = number) OR (flag = TRUE)
16  END IF
17
18  IF NOT (flag = TRUE)
19     SEND number TO DISPLAY
20  END IF
21
```

**Figure 3**

Trace tables are used to show the outputs from algorithms.

(i) Complete the trace table for an input of 7

**(2)**

| number | flag | testNum | DISPLAY |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
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| | | | |

## 1. Question 6 · part a · 6 marks · Working and answer

Working and answer

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## 2. Question 5 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

**5** Ships carry cargo around the world in containers.

(a) Containers come in two sizes.

**Figure 2** shows an algorithm written using flowchart symbols.

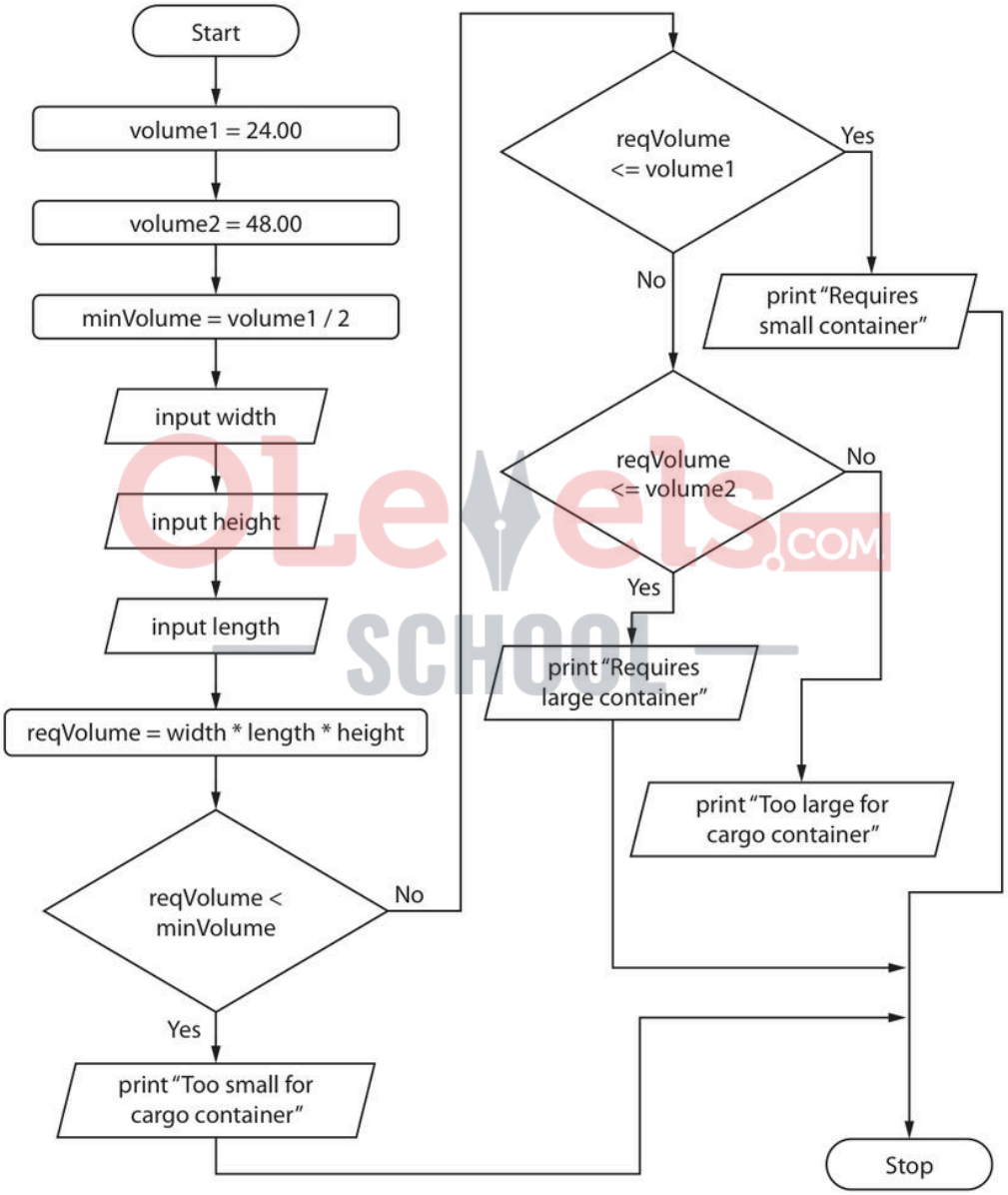



**Figure 2**

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## 2. Question 5 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

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(i) Complete the table to show the output for each cargo item.

(3)

| Cargo item | | | Output |
|---|---|---|---|
| width | length | height | |
| 4 | 4 | 2 | |
| 2 | 2 | 2 | |
| 3 | 8 | 5 | |

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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## 2. Question 5 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(b) Each ship is registered to a state.

**Figure 3** shows a partially completed algorithm written in pseudocode.

The completed algorithm must:

- print each state to the display on a new line
- count the number of states
- create a message as a single string (e.g. there are *number* states)
- print the message to the display.

Complete the algorithm in the space provided in **Figure 3**.

(4)

```
SET numStates TO 0
SET states TO ["France", "Singapore", "Malta", "Panama", "Greece", "Italy"]

FOR EACH state FROM states DO




END FOREACH
```

**Figure 3**

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P 6 1 8 8 0 A 0 1 6 2 4

## 2. Question 5 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

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## 2. Question 5 · part b · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part b only.

(c) Cargo ships have maximum weight loads.

**Figure 4** shows an algorithm written using pseudocode.

The algorithm should identify the size of cargo ship required for any load.

There is an error on line 9.

```
2   SET loadWeight TO [20000, 28000, 40000, 50000]
3   SET index TO 0
4   SET found TO FALSE
5
6   SEND "Enter cargo weight" TO DISPLAY
7   RECEIVE target FROM (INTEGER) KEYBOARD
8
9   WHILE (NOT found) DO
10      IF (loadWeight [index] >= target) THEN
11          SEND loadWeight [index] TO DISPLAY
12          SET found TO TRUE
13      ELSE
14          SET index TO  index + 1
15      END IF
16  END WHILE
17
18  IF (NOT found) THEN
19      SEND "No ship available" TO DISPLAY
20  END IF
```

**Figure 4**

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## 2. Question 5 · part b · 4 marks · Working and answer

Working and answer

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## 3. Question 6 · part a · 6 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

**6** A software engineer is working with some algorithms.

(a) An algorithm needs to be developed.

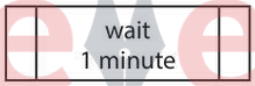
The algorithm must:

- check for a change of hour every minute
- use the library subprogram getHour() to get the hour part of the current time in the 24-hour clock (0 to 23)
- output "Good morning" when the hour is between 3 and 12, inclusive
- output "Good afternoon" when the hour is between 13 and 19, inclusive
- output "Good night" at all other times.

Complete the flowchart to represent this algorithm, in the space provided on the next page.

The call to the library subprogram getHour() has already been included.

Use this subprogram symbol to show waiting for 1 minute.




**(6)**

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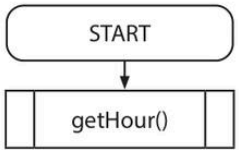
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## 3. Question 6 · part a · 6 marks

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Further practice · Attempt part a only.

Draw your flowchart here.




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## 3. Question 6 · part a · 6 marks

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Further practice · Attempt part a only.

(b) Another algorithm determines when to change the flavoured syrups for a drinks dispensing machine.

**Figure 4** shows this algorithm written in pseudocode.

```
1  SET flavours TO ["anise", "mango", "cola", "apple", "papaya", "strawberry",
   "lychee", "banana"]
2
3  SET volume TO [0.7, 0.2, 0.6, 0.1, 0.05, 0.8, 0.4, 0.6]
4
5  SET i TO 0
6
7  WHILE i < 8 DO
8
9      IF (volume[i] < 0.1) THEN
10         SEND (flavours[i] & " needs changing") TO DISPLAY
11
12     ELSE
13         IF (volume[i] >= 0.3) AND (volume[i] <= 0.5) THEN
14             SEND (flavours[i] & " needs ordering")  TO DISPLAY
15         END IF
16     END IF
17
18     SET i TO i + 1
19
20 END WHILE
```

**Figure 4**

(i) Give the output produced by the algorithm.

**(2)**

The algorithm works with a fixed number of flavours.

(ii) Give the number of the line in the algorithm that would need to be amended to allow for any number of flavours.

**(1)**

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P 6 1 8 8 4 R R A 0 1 8 2 0

## 3. Question 6 · part a · 6 marks · Working and answer

Working and answer

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# 4. Question 4 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

**4** Algorithms can be used to perform calculations and to process data.

(a) State what is meant by the term **algorithm**.

**(1)**

(b) **Figure 3** shows an algorithm.

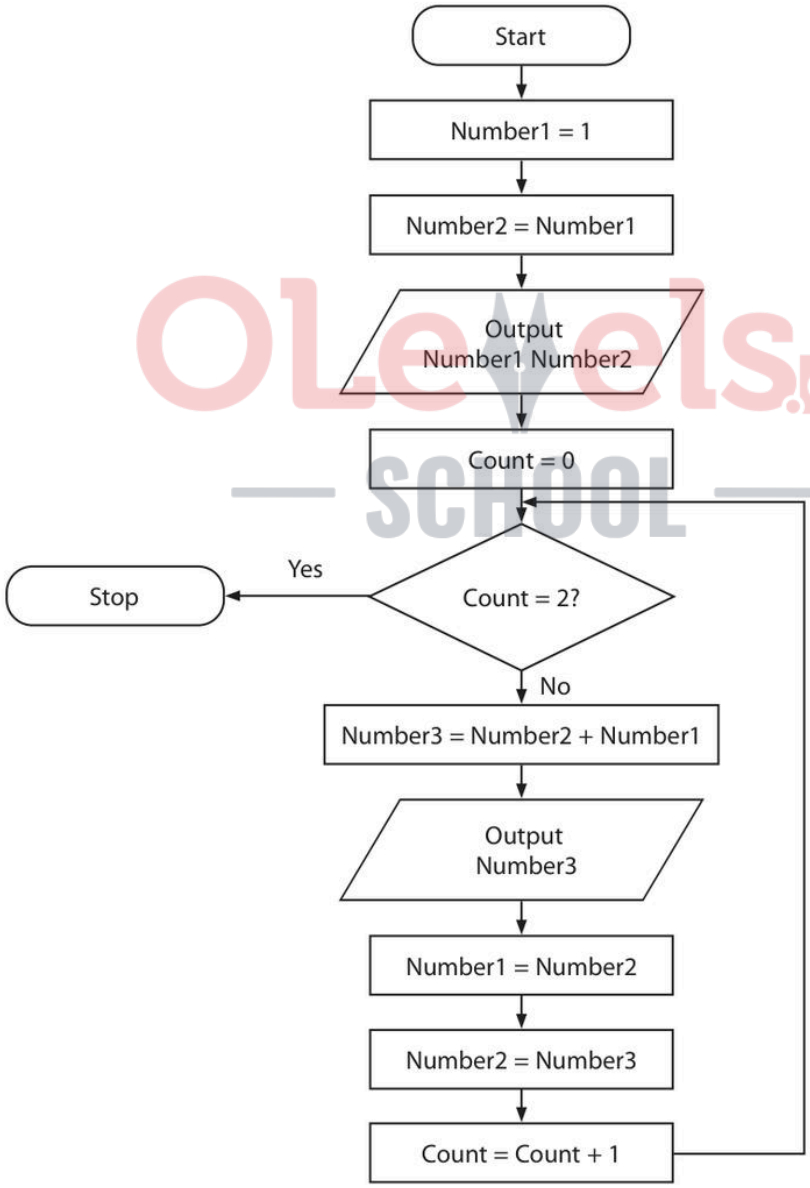



**Figure 3**

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# 4. Question 4 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

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(i) Complete this trace table for the algorithm.

You may not need to use all the rows.

(5)

| Number1 | Number2 | Number3 | Count | Output | Count = 2? |
|---|---|---|---|---|---|
| 1 | 1 | – | 0 | 1 1 | False |
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |

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(ii) The benefits of using a trace table include that they allow variable states, outputs and decisions to be recorded.

Give **one other** benefit of using a trace table to test an algorithm.

(1)

.....

.....

P 6 9 2 9 7 A 0 1 3 2 0

# 4. Question 4 · part c · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part c only.

(c) An algorithm is needed to count and display the number of vowels in a word.

The vowels are a, e, i, o, and u.

The completed algorithm must:

- count the number of vowels
- create a message as a single string (e.g. there are *number* vowels)
- print the message to the display.

Here is a partially completed algorithm written in pseudocode.

Complete the algorithm in the spaces provided.

**(5)**

```
SET word TO 'elephant'

FOR EACH letter FROM word DO

END FOREACH
```

**(Total for Question 4 = 12 marks)**

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## 4. Question 4 · part c · 5 marks · Working and answer

Working and answer

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## 5. Question 4 · part a.i · 5 marks

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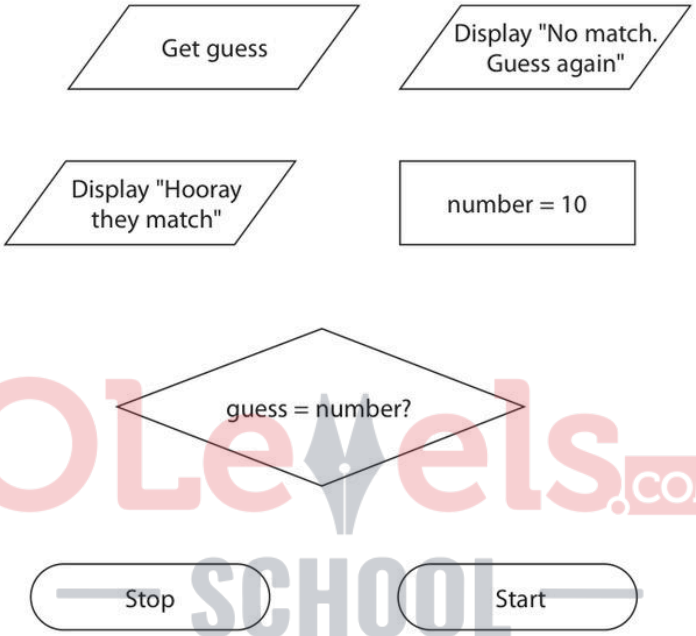
Further practice · Attempt part a.i only.

**4** Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.




Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

**(5)**

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## 5. Question 4 · part a.i · 5 marks

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Further practice · Attempt part a.i only.

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**Draw your flowchart here.**

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## 5. Question 4 · part a.i · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part a.i only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

| | Step |
|---|---|
| A | Change the temperature to Celsius |
| B | Get the temperature in Fahrenheit |
| C | Show the temperature in Celsius |
| D | Set the temperature to 0 |

(i) Give the letter of the step that initialises a variable.

(1)

.....

(ii) Give the letter of the step that inputs a value.

(1)

.....

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## 5. Question 4 · part a.i · 5 marks · Working and answer

Working and answer

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# 6. Question 6 · part d · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part d only.

**6** Pseudocode and flowcharts are used to create algorithms.

(a) The number of minutes a train is early or late are recorded each day for a week.

- A value of 0 is recorded if the train is on time.
- A negative value is recorded if the train is early.
- A positive value is recorded if the train is late.

The algorithm in **Figure 2** has been created to calculate and output the total number of trains that were early, on time or late.

```
01
02  SET arrival TO [-2, 1, 5, 0, -3, 4, 1]
03
04  SET early TO 0
05  SET late TO 0
06  SET index TO 0
07
08  WHILE NOT (index > LENGTH(arrival)) DO
09
10      IF arrival[index] >= 0 THEN
11          SET late TO arrival[index]
12      ELSE
13          IF arrival[index] < 0 THEN
14              SET early TO early + 1
15          END IF
16      END IF
17
18      SET index TO index + 1
19
20  END WHILE
21
22  SEND "Trains early:    " & early TO DISPLAY
23  SEND "Trains on time: " & (late + early) TO DISPLAY
24  SEND "Trains late:     " & late TO DISPLAY
25
```

**Figure 2**

Identify the type of iteration construct used in **Figure 2**.

(1)

- ☒ **A** Post-conditioned loop
- ☒ **B** Count controlled loop
- ☒ **C** Array loop
- ☒ **D** Pre-conditioned loop

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P 7 5 7 3 6 A 0 2 2 2 8

## 6. Question 6 · part d · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part d only.

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(b) There are **four** errors in the pseudocode in **Figure 2**.

Write the correct code for lines 08, 10, 11 and 23

(i) Line 08

**(1)**

(ii) Line 10

**(1)**

(iii) Line 11

**(1)**

(iv) Line 23

**(1)**

(c) Give **one** reason for the variable `index` being incremented in line 18.

**(1)**

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# 6. Question 6 · part d · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part d only.

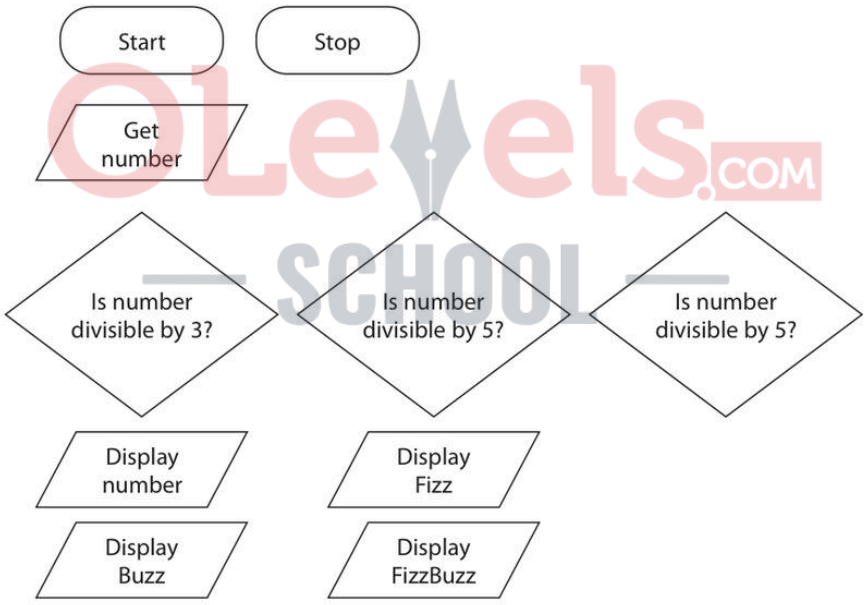
(d) In the game FizzBuzz a user enters a number.

These conditions apply:

- if the number is divisible by 3 the program displays Fizz
- if the number is divisible by 5 the program displays Buzz
- if the number is divisible by both 3 and 5 the program displays FizzBuzz
- if the number is not divisible by either 3 or 5, the program displays the number entered.

Hint: If one number is divisible by another number there is no remainder.

These are the components needed for the flowchart.




Draw the flowchart for the algorithm used in the game in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(6)

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## 6. Question 6 · part d · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part d only.

**Draw your flowchart here.**

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**(Total for Question 6 = 12 marks)**

## 6. Question 6 · part d · 6 marks · Working and answer

Working and answer

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